Mapping Collaborations in Performing Arts: Building the Festival d'Avignon Digital Corpus

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This poster outlines the creation of the pAFA corpus, a digital resource built from the show programmes ("programme de salle" in French, is a document provided to audience members on the day of the performance. It offers key information about it — from production credits and biographical notes on the author or director to any other details that help contextualize and enhance the audience's understanding of the show.) of the Festival d'Avignon (1947–2024). As part of the ERC-funded STAGE project, this corpus supports the study of collaborations in contemporary European performing arts.

Context

STAGE aims to develop a new historiography of mise en scène and creative processes in post-WWII Europe by integrating performing arts studies with digital tools, particularly data-driven AI methods. One goal is to examine creation contexts and networks through the lens of collaboration. Building on actor-network theory (Latour 1999; 2005) and the works of Caplan (2016), Miller (2014), and Bardiot (2021), we hypothesise that mapping networks of performance contributors will offer new insights into the intricate contexts of contemporary European stagings. Visualising these networks will reveal patterns and influences, highlighting the collaborative nature of performances and the interplay among diverse artists across disciplines, languages, and cultures.

By focusing on show programmes – a valuable yet understudied resource – we seek to extract, structure, and analyse information about contributors, professions, and collaborations. A show programme often includes the place, date, distribution, photographs and text of the related performance. Information found in show programmes tends to be more reliable and up to date than in season programmes, since the former programmes are printed very close to the premiere performance date, and not a year in advance like season programmes. Moreover, existing databases (Roussillon & Schuwey 2020) often reduce show programmes to minimal metadata, overlooking the rich contextual data available in their literal transcription. In contrast, our approach preserves the original content of the document while structuring it for computational analysis.

As a starting point, we are examining a multimodal French corpus from the Archives of the Festival d'Avignon (AFA), including 1,870 show programmes in PDF format. This major event, featuring over 2,000 performances since 1947, provides an unprecedented resource for performing arts analytics.

Corpus creation

We built a corpus called the pAFA corpus, by processing the 1,870 AFA's programmes aforementioned, representing a total of 8,139 pages. This corpus will enable us to map out networks of Avignon's contributors in the performing arts, based in particular on names and corresponding professions. In this poster, we detail the pAFA corpus construction.

The main steps for creating the pAFA corpus are: (a) collection, (b) processing, and (c) release.

Collection (a)

The corpus was compiled from two sources: the Bibliothèque nationale de France (BnF) through Gallica, and the Festival d'Avignon's public relations service (Avignon). Differences in data organisation required harmonisation before merging them into a unified corpus. The corpus covers 1947 to 2024, with born-digital documents for recent years (since 2003) and paper archives for earlier ones. The digitisation process was necessary for older documents (before 2003). A naming system identifies the source, name, and date for each PDF document (e.g. Gallica_prog_FDA2015.pdf). We used the ShareDocs cloud service by IR* HumaNum for data storage.

Processing (b)

Processing is automated: (i) preprocessing original documents (segmentation and OCR), (ii) parsing preprocessed documents, and (iii) extracting content from the parsed representations obtained (text, tables and images), ideally without any information loss, particularly preserving words initial forms (i.e. the literal transcription of the text found in original documents).

Preprocessing : all PDFs are split in sub-pages using an in-house PDF splitter implemented in Python (based on dhSegment (Ares Oliveira et al. 2018)), as it helps to read text in the correct order for parsing documents. Pre-2003 PDFs, containing only scanned images, received an additional text layer using a reliable OCR solution (Kiessling 2021).

Parsing and extracting : we use IBM's PDF content-based open-source Python library Docling (Livathinos et al. 2025) for parsing preprocessed documents. IBM's solution is state- of-the-art, actively updated, and shown to be much more efficient than any other solutions we tested such as pdfminer.six , pyMuPDF , unstructured.io , and minerU (Wang et al. 2024).

The corpus will be available in two formats, human and machine-readable respectively : (f1) markdown and (f2) DoclingDocument (markdown in JSON format with PDFs coordinates), both formats suitable for LLM ingestion, along with IIIF link to original PDF for each document. DoclingDocument is Docling's unified document representation. It can be enriched and used by downstream applications (such as LangChain and Spacy) through IBM's API to inspect, export, or chunk documents for various purposes. As Pydantic data model, this format can express various common document features.

Release (c)

This step involves both manual and automatic controls of the processing results to assess the quality of the corpus. The pAFA corpus and related extraction tools will be made accessible on Nakala, a platform supported by IR* HumaNum.

Perspectives

The pAFA corpus will facilitate analysis of technical and artistic roles as well as their related relationships and evolution over time by mapping networks of performances participants.

We also plan to ingest AFA's programmes into a multi-modal RAG system — an innovative approach that combines various data types, such as text, images and videos, using Colpali (Faysse et al. 2025) through Byaldi . This way, we will be able to evaluate the relevance of historiographical studies through natural language as well as to understand whether the related LLM-generated Knowledge Graph might help us in developing our performing arts ontology (Bardiot et al. 2025).

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